



Business Requirements Document (BRD)

AI-Powered Lebanon Trip & Experience Platform

Product promise

Discover. Plan. Book Lebanon. Mshwar turns a user's intent, budget, time and preferences into a feasible, personalized itinerary built from real businesses, activities and attractions - with availability and booking workflows.

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Product Scope	Lebanon-wide consumer platform (separate from the Tripoli-specific project)
Prepared For	Mshwar Project Team / Academic & Product Review
Status	Ready for product design, FRD/SRS, architecture and implementation planning

Confidentiality: Project planning document - not a legal or financial commitment.

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1. Document Purpose & Product Definition

This BRD defines the business objectives, scope, stakeholders, marketplace model, user journeys, functional requirements, AI requirements, data needs, non-functional requirements, controls, risks and acceptance criteria for Mshwar. It is the baseline for UX design, FRD/SRS creation, architecture, implementation, testing and launch planning.

Scope clarification

Mshwar is a Lebanon-wide product. The previously discussed Tripoli-specific concept is treated as a separate project and is not used as the geographic scope of this BRD.

Mshwar is not defined as a generic travel chatbot. The product must generate plans from verified structured platform data and convert recommendations into discoverable, reservable or bookable experiences. AI is an enabling layer; the business product is the marketplace and booking workflow.

1.1 Product Definition

Field	Definition
Product Name	Mshwar
Category	Travel, local discovery, experience marketplace and itinerary planning platform
Primary Geography	Lebanon
Primary Users	Residents, domestic travelers, tourists, couples, families and friend groups
Supply Users	Restaurants, experience providers, guesthouses, resorts, tour operators, attractions and activity businesses
Core Promise	Turn a user request into a feasible itinerary using real places, prices, timing and availability, then support booking/reservation.
Languages	Arabic, English and French; Lebanese Arabic understood in free-text trip requests
Primary Channels	Responsive web application; mobile-first UX

1.2 Confirmed Product Decisions

Decision	Baseline
Database	PostgreSQL + PostGIS + pgvector; MongoDB is not the baseline because relational, geospatial and booking integrity requirements are central.
Frontend	Next.js + React + TypeScript + Tailwind CSS + shadcn/ui
Backend	FastAPI + Python + SQLAlchemy + Pydantic + Alembic
AI Pattern	LLM + RAG for language/explanations, structured database retrieval for facts, deterministic validation and OR-Tools for hard planning constraints
Maps	Google Maps Platform
Weather	Open-Meteo for prototype; commercial/self-hosted option required for production use
Images	ImageKit
Payments	Stripe in test/demo flows; production payment provider must be abstracted because Lebanon is not currently listed as a supported Stripe country
Monitoring	Sentry
Deployment	Vercel for frontend; Supabase/PostgreSQL for data; backend hosting selected during technical design

2. Executive Summary

Planning a day out or short trip in Lebanon is fragmented. Users typically move between social media, maps, search, phone calls and messaging to discover places, compare options, estimate driving time, check opening hours and confirm reservations. The information required to make a good plan is distributed across many sources and is often not structured for itinerary creation.

Mshwar centralizes this journey. A user can either browse normally or describe a desired outing in natural language. The platform extracts preferences, retrieves verified candidate experiences, evaluates budget, travel time, opening hours, group size, availability and weather, optimizes the sequence of stops, and presents an itinerary that the user can edit, share and book.

The supply side gives participating businesses a portal to manage their listings, experiences, prices, capacity, availability, booking rules and reservation requests. Mshwar can monetize demand generation through commissions, business subscriptions, qualified leads and clearly labeled promoted placement.

North-star outcome

A user should be able to go from "what should we do?" to a realistic, bookable plan without manually coordinating multiple disconnected websites and businesses.

3. Business Context & Problem Statement

3.1 Customer Problems

- Discovery is fragmented across maps, search engines, social media pages and personal recommendations.

- A recommendation can be attractive but infeasible because of distance, timing, opening hours, capacity, weather or budget.
- Groups struggle to agree on preferences and budget, causing repeated manual planning.
- Users cannot easily compare experiences on one standardized set of attributes such as price, duration, availability and cancellation rules.
- Generic AI tools may invent businesses, outdated prices or impossible schedules because they are not connected to authoritative marketplace data.

3.2 Business Problems

- Local experience businesses often depend on social media and messaging for discovery and reservations.
- Small providers have limited visibility into demand, conversion and booking performance.
- Businesses repeatedly answer the same questions about price, availability, location and group suitability.
- There is no shared planning layer that places businesses inside complete, contextual itineraries rather than isolated search results.

3.3 Opportunity

Mshwar can create value by combining marketplace inventory with itinerary intelligence. The defensible product is the continuously improving supply database, availability, booking history, user preference signals and itinerary outcome data - not the LLM alone.

4. Product Vision, Value Proposition & Principles

4.1 Vision

Vision statement

Become the default digital layer for discovering, planning and booking experiences across Lebanon - from a spontaneous afternoon to a full-day trip.

4.2 Value Proposition

Audience	Value
Travelers / Locals	One place to discover, plan, optimize, share and book a day or trip.
Groups	Shared planning, voting and budget-aware suggestions that reduce coordination friction.
Businesses	Qualified customer demand, bookings, visibility and performance insights.
Platform	Marketplace revenue plus proprietary demand, preference and conversion data.

4.3 Product Principles

- Real before clever: never prefer an impressive AI answer over verified availability, price and feasibility.
- Structured facts are authoritative: the LLM must not override database truth.
- Bookability matters: recommendations should lead to a real action wherever possible.
- Explainable planning: users should understand why an experience was selected and what constraints shaped the plan.

- User control: every AI-generated itinerary must be editable; users can replace or lock stops.
- Trust and transparency: sponsored placement, uncertain availability and estimated prices must be clearly labeled.
- Mobile-first and multilingual: the product should work naturally for users switching between Arabic, English and French.

5. Business Objectives & Success Metrics

ID	Objective	Business Outcome	Primary KPI
BO-01	Reduce planning friction	Enable users to create a feasible itinerary in minutes rather than manually coordinating multiple sources.	Trip-builder completion rate; median time to itinerary
BO-02	Generate marketplace transactions	Convert discovery and planning into reservation or booking actions.	Itinerary-to-booking conversion; confirmed bookings
BO-03	Create business value	Deliver qualified demand to participating businesses.	Active businesses; booking value; business response/confirmation rate
BO-04	Build trusted inventory	Maintain accurate places, prices, policies and availability.	Verified listing coverage; stale-data rate; dispute rate
BO-05	Improve recommendations over time	Learn from edits, saves, bookings and reviews.	Acceptance rate; replacement rate; repeat planning rate
BO-06	Operate safely and reliably	Protect users, businesses and platform integrity.	Payment/booking error rate; security incidents; uptime; support resolution time

5.1 Initial KPI Targets for MVP Validation

The following are validation targets, not contractual guarantees. They should be recalibrated after pilot data is available.

Metric	Initial Target / Guardrail
Itinerary validity	100% of displayed stops must map to known platform entities; no invented businesses.
Hard-constraint violations	0 unflagged violations for budget, opening hours, capacity or required return time after validation.
Trip builder completion	Track funnel; target >40% after users submit a sufficiently complete request.
Business confirmation	Target >70% for inventory configured as request-to-book during pilot.
AI response failure	<2% unrecovered AI failures during normal operation; non-AI fallback available.
Page performance	Core interactive pages should be usable on common mobile connections; target p75 LCP <=2.5s where technically feasible.

6. Scope & Release Boundaries

6.1 In Scope

- Consumer accounts, profiles, preferences and saved trips.
- Public browse/search across destinations, businesses, activities, attractions and ready-made trips.
- Natural-language AI trip builder with budget, time, group and interest constraints.

- Interactive map, route/time calculations and itinerary sequencing.
- Weather-aware warnings and optional replanning for weather-sensitive activities.
- Availability representation, request-to-book and instant-confirm modes where supported.
- Reservation/booking lifecycle, payment abstraction and Stripe test-mode integration for the academic/demo environment.
- Group trip sharing and voting.
- Favorites, reviews and ratings.
- Business onboarding, verification, listing/experience management, availability and booking dashboard.
- Admin moderation, verification, support, analytics and configuration.
- Arabic, English and French UI; Lebanese Arabic handling in AI input.
- RAG over verified business/experience knowledge plus structured retrieval and deterministic validation.

6.2 Explicitly Out of Scope for MVP

- Full airline, hotel GDS or international travel booking.
- Owning or operating transportation fleets.
- Guaranteed live availability from businesses that do not maintain authoritative inventory.
- Automated legal, visa or safety advice.
- AI-generated reviews or fabricated popularity signals.
- Unverified web scraping used as authoritative price or availability data.
- Native iOS/Android applications; the MVP is responsive web first.
- Complex loyalty points, gift cards, BNPL, crypto or multi-currency settlement.

6.3 MVP vs Later Scope

Capability	MVP	Later
Trip Builder	Natural language + filters + validated itinerary	Deep personalization, collaborative ranking, automatic rebooking suggestions
Booking	Request-to-book + instant confirm where reliable	Broader real-time inventory integrations
Payments	Stripe test/demo + provider abstraction	Production PSP/local acquiring, refunds, marketplace payouts as legally/operationally supported
Group Planning	Share link + voting + participant preferences	Real-time collaborative editing and split payments
Businesses	Listing, experience, slots, booking inbox, analytics	Promotions, dynamic offers, API integrations
AI	Intent extraction, ranking, itinerary explanation/revision	Learned ranking, richer personalization, demand forecasting

7. Stakeholders, Roles & Personas

7.1 Stakeholders

Stakeholder	Primary Interest	Responsibilities
Product Owner	Product-market fit, scope and business outcomes	Prioritize roadmap, approve requirements, resolve trade-offs
Travelers / Users	Relevant plans, transparent pricing, easy booking	Provide preferences, verify booking details, review experiences
Business Owners	Demand, bookings, manageable operations	Maintain listing data, prices, availability and policies
Business Staff	Day-to-day booking operations	Confirm/reject requests, update slots, communicate changes
Platform Admin	Trust, supply quality, support and policy enforcement	Verify businesses, moderate content, manage disputes/configuration
Engineering	Reliable, secure, maintainable system	Implement, test, deploy, observe and support platform
AI/Data	Valid, useful recommendations	Evaluation, retrieval, ranking, safeguards and feedback loop

7.2 Primary Personas

Persona	Context	Top Needs
P1 - Friend Group Planner	Organizes outings for 3-8 people; wants budget clarity and consensus.	Fast group plan, activity variety, route efficiency, shareable itinerary.
P2 - Couple	Wants romantic or premium experiences with minimal planning effort.	Atmosphere, timing, privacy, dinner/activity combinations, easy reservation.
P3 - Family	Needs age suitability, predictable timing, parking/accessibility and budget control.	Family-friendly filters, duration, facilities, weather backup.
P4 - Tourist	Has limited local knowledge and needs reliable, multilingual guidance.	Verified places, map context, clear policies, booking confidence.
P5 - Experience Business	Needs more qualified customers without complex software.	Easy listing, availability, booking inbox, analytics and transparent fees.
P6 - Platform Admin	Protects marketplace quality and resolves operational issues.	Verification, moderation, auditability, overrides and reporting.

8. Business Model & Marketplace Design

8.1 Marketplace Structure

Mshwar is a two-sided marketplace. Demand-side value comes from planning and booking convenience. Supply-side value comes from qualified discovery and bookings. The platform should support different business models without tightly coupling them to the booking engine.

Revenue Stream	Mechanism	MVP Position
Booking Commission	Configurable percentage or fixed fee on completed bookings.	Supported by data model; commercial rate validated with partners.

Revenue Stream	Mechanism	MVP Position
Business Subscription	Monthly plan for enhanced visibility, analytics, inventory tools or lower commission.	Later pilot / configurable.
Qualified Lead Fee	Fee for high-intent requests where payment happens off-platform.	Optional for categories without direct checkout.
Promoted Placement	Sponsored ranking slots clearly labeled as sponsored.	Later; must not silently override organic relevance.
Premium User Features	Advanced planning, saved preferences or exclusive benefits.	Future, not MVP dependency.

8.2 Supply Quality Model

- Listings must have an owner, category, location, status and verification state.
- Prices must be marked as fixed, from-price, range, per-person, per-group or quote-required.
- Availability must record its source and last-updated time.
- Verified businesses receive a visible trust indicator after administrative review.
- User-facing ranking may consider quality and reliability, but paid placement must be explicitly labeled.

9. End-to-End User Journeys

J1 - AI Plan & Book

1. User enters start location, date, group size, budget, available time and free-text preferences.
2. AI extracts structured intent and asks only for missing mandatory constraints.
3. Backend retrieves verified candidates and filters infeasible options.
4. Optimizer sequences stops against travel time, opening hours, duration, capacity and budget.
5. User reviews map, total cost, timing and rationale; locks/replaces stops if desired.
6. User books/reserves eligible items; confirmations appear in one itinerary.

J2 - Browse & Build Manually

7. User browses destinations/experiences using search, filters and map.
8. User saves items to a trip.
9. System suggests compatible nearby additions and highlights conflicts.
10. User sequences the trip and books selected experiences.

J3 - Group Planning

11. Organizer creates a shared trip and invites participants by link.
12. Participants vote on categories/places and optionally add budget/preferences.
13. Mshwar calculates a group-fit recommendation and shows trade-offs.
14. Organizer finalizes and books the itinerary.

J4 - Weather Replan

15. System detects that an outdoor stop may be affected by forecast conditions.
16. User receives a clear warning, not an automatic silent change.
17. User chooses Rebuild Affected Stops.
18. System substitutes compatible indoor/weather-safe options while preserving locked stops, budget and return time.

J5 - Business Onboarding

19. Business creates account and submits identity/contact/listing data.
20. Admin verifies business before verified status is granted.
21. Business creates experiences, pricing, availability and booking rules.
22. Business receives booking requests/instant bookings and manages status.

J6 - Booking Exception

23. Business cancels or cannot fulfill a reservation.
24. System informs affected user, records reason and refund state if applicable.
25. Mshwar suggests validated alternatives that preserve time/location constraints.
26. Admin can intervene if a dispute or refund exception occurs.

10. Functional Requirements

Priorities use MoSCoW: Must (M), Should (S), Could (C). Requirements are written at business/functional level; detailed API contracts and UI specifications belong in the FRD/SRS and design system.

10.1 Account, Profile & Preferences

ID	Business / Functional Requirement	Priority	Acceptance / Business Rule
FR-001	Users can register, sign in, sign out and recover access using secure authentication.	M	Protected actions require authenticated session; failed authentication does not expose account existence beyond safe UX.
FR-002	Users can maintain profile basics, preferred language, home/start area and optional travel preferences.	M	Preferences are editable and reflected in future planning only when consented/appropriate.
FR-003	Users can save dietary, accessibility, activity intensity and interest preferences.	S	Preferences can be overridden per trip; sensitive preferences are not inferred without user input.
FR-004	Users can view and manage saved trips, favorites and booking history.	M	Items are scoped to the authenticated user and support deletion where policy permits.
FR-005	Users can choose Arabic, English or French interface language.	M	Arabic UI supports RTL; language can be changed without creating a new account.

10.2 Discovery, Search & Listings

ID	Business / Functional Requirement	Priority	Acceptance / Business Rule
FR-010	Users can browse destinations, experiences, attractions, restaurants and ready-made trips.	M	Only active/published content is visible publicly.
FR-011	Search supports keyword, category, location and natural-language discovery.	M	Results identify source entities and never display fabricated listings.
FR-012	Filters include date, price/budget, distance, category, group suitability, rating and availability where known.	M	Filter state is reflected in result set and shareable URL where practical.
FR-013	Users can switch between list and interactive map views.	M	Map markers correspond to visible result entities and open the correct detail card.
FR-014	Listing pages show verified business/experience data, price model, location, duration, policies, images and availability status.	M	Estimated/from-price/quote-required values are explicitly labeled.
FR-015	Users can favorite listings and add them to a trip.	M	Duplicate favorites are prevented and state syncs across sessions.
FR-016	The platform can surface related/nearby experiences without forcing AI trip generation.	S	Suggestions are based on location/category/compatibility and include transparent distance/time context.

10.3 AI Personalized Trip Builder

ID	Business / Functional Requirement	Priority	Acceptance / Business Rule
FR-020	The user can describe a desired trip in Arabic, Lebanese Arabic, English, French or mixed language.	M	Intent parser returns validated structured fields or asks for missing required inputs.
FR-021	The trip builder captures date, start point, group size, budget, time window and interests as core constraints.	M	A plan is not finalized until required constraints are known or explicitly defaulted with user-visible assumptions.
FR-022	The system retrieves candidate places from Mshwar inventory before generating itinerary text.	M	Every recommended stop has a valid internal entity ID and active listing status.
FR-023	The system filters candidates against opening hours, duration, capacity/availability, travel feasibility and budget.	M	Hard constraint violations are blocked or clearly flagged before display.
FR-024	The system optimizes the order of stops to reduce unnecessary travel while respecting user priorities.	M	Route/time values are calculated by mapping/routing services and optimizer output is validated.
FR-025	The generated itinerary displays timeline, estimated total cost, travel segments, stop durations and booking status.	M	Total is computed from structured values, not generated free text.
FR-026	Users can lock a stop and request changes to other parts of the itinerary.	M	Locked items remain unchanged unless a feasibility conflict makes them impossible; conflict must be explained.
FR-027	Users can replace a specific stop with alternatives matching remaining constraints.	M	Replacement recalculates downstream time, cost and route before confirmation.
FR-028	Users can conversationally refine a trip, e.g. "less driving" or "make it more romantic".	S	Free-text edits are converted into structured preference changes and revalidated.
FR-029	The system explains key recommendation reasons without exposing private model internals.	S	Explanations reference user preferences and factual attributes only.

10.4 Maps, Routing & Weather

ID	Business / Functional Requirement	Priority	Acceptance / Business Rule
FR-030	Users can choose start location by search, map selection or permitted device location.	M	Precise device location is requested only with user permission and can be replaced manually.
FR-031	The itinerary displays map route, travel time and distance between stops.	M	Displayed route metrics come from the configured routing provider or are marked unavailable.
FR-032	The system uses route optimization to sequence multi-stop itineraries.	M	Optimization respects locked order, appointment times and closing/return-time constraints.
FR-033	Weather-sensitive activities carry an indoor/outdoor/weather-sensitivity classification.	M	Classification is editable by verified business/admin and used by planning rules.
FR-034	The system warns when forecast conditions may affect planned weather-sensitive stops.	M	Warning includes affected stops and forecast source timestamp; it does not silently cancel bookings.
FR-035	Users can rebuild only weather-affected portions of a trip.	S	Locked and unaffected stops remain fixed where feasible.

10.5 Availability, Booking & Payments

ID	Business / Functional Requirement	Priority	Acceptance / Business Rule
FR-040	Businesses can configure experiences as instant-confirm, request-to-book or inquiry-only.	M	User sees the booking mode before committing.
FR-041	Availability supports date/time slots, capacity and blackout periods.	M	Overbooking is prevented for authoritative inventory using transactional checks/locking.
FR-042	Users can reserve/book an experience from listing or itinerary context.	M	Booking captures party size, date/time, price snapshot, policy snapshot and status.
FR-043	The system maintains booking states: draft, pending, confirmed, rejected, cancelled, completed and refunded/partially refunded where applicable.	M	State transitions are audited and invalid transitions are blocked.
FR-044	Payment requests are created only after backend validation of price, availability and booking status.	M	The LLM cannot initiate or alter financial amounts directly.
FR-045	Payment processing uses an abstract provider interface; Stripe test mode is supported for demo/testing.	M	Provider-specific IDs are stored separately from business booking IDs; webhooks are verified.
FR-046	Users receive confirmation with itemized booking details and policies.	M	Confirmation is generated from persisted booking/payment data.
FR-047	Cancellation/refund eligibility is computed from the policy snapshot attached to the booking.	M	Policy changes after booking do not retroactively change that booking unless legally required.
FR-048	Failed or timed-out payment attempts must not create a confirmed paid booking.	M	Payment and booking writes are idempotent and reconcilable.

10.6 Group Planning, Favorites & Reviews

ID	Business / Functional Requirement	Priority	Acceptance / Business Rule
FR-050	A trip owner can create a shareable group-planning link.	M	Link permissions distinguish view, vote and edit roles.
FR-051	Participants can vote on suggested experiences/categories and optionally submit budget/preferences.	M	Votes are attributed to participant/session and can be updated until trip lock.
FR-052	The group recommendation engine summarizes agreement and major trade-offs.	S	No participant private preference is shown beyond what they agreed to share.
FR-053	Users can save/favorite trips and individual listings.	M	Favorites survive sessions and are removable.
FR-054	Users can review/rate eligible businesses or experiences after a relevant completed/confirmed interaction.	M	Review eligibility is enforced and duplicate abuse is mitigated.
FR-055	Businesses can respond to reviews but cannot edit or suppress legitimate reviews.	S	Responses are moderated under platform policy.

10.7 Business Portal

ID	Business / Functional Requirement	Priority	Acceptance / Business Rule
FR-060	Businesses can create an organization account and submit verification information.	M	Verification state is pending until admin approval.
FR-061	Business owners can invite staff with role-based permissions.	S	Roles limit access to listings, bookings, finance and settings.
FR-062	Businesses can create/edit listings and experiences with categories, images, location, duration, pricing and policies.	M	Required fields and data types are validated before publication.
FR-063	Businesses can manage opening hours, exceptions, slots, capacity and blackout dates.	M	Changes record actor and timestamp.
FR-064	Businesses can confirm/reject request-to-book reservations with reason and optional message.	M	Response updates user itinerary and triggers notification.
FR-065	Businesses can cancel bookings according to policy and operational permissions.	M	Cancellation requires reason; affected payment/refund workflow is triggered.
FR-066	Businesses can view booking calendar/list and filter by status/date/experience.	M	Only organization-authorized bookings are visible.
FR-067	Businesses can view performance metrics such as views, saves, itinerary inclusions, booking requests, confirmations and revenue where applicable.	S	Metrics are based on auditable platform events.
FR-068	Businesses can mark content temporarily unavailable without deleting historical records.	M	Unpublished/paused inventory is excluded from new itineraries.
FR-069	Businesses can define contact/support details and fulfillment instructions.	M	Sensitive internal contact data can be separated from public contact data.

10.8 Admin, Moderation & Support

ID	Business / Functional Requirement	Priority	Acceptance / Business Rule
FR-070	Admins can review, approve, reject, suspend and re-verify businesses.	M	Actions require reason and are audit logged.
FR-071	Admins can moderate listings, images, reviews and user-generated content.	M	Moderation preserves evidence/history where required for audit/support.
FR-072	Admins can manage categories, tags, amenities, suitability labels and weather-sensitivity taxonomy.	M	Changes propagate to search/planning without breaking existing records.
FR-073	Admins can inspect booking/payment state and perform authorized support actions.	M	Sensitive actions require elevated permission and audit trail.
FR-074	Admins can configure fees/commission, feature flags and marketplace rules without code changes where appropriate.	S	Configuration changes are versioned/audited.
FR-075	Admins can view platform health and business/user KPIs.	M	Dashboard metrics use defined event semantics and date filters.
FR-076	Admins can handle reported reviews/listings and user support cases.	S	Case status, owner, evidence and resolution are tracked.
FR-077	Admins can trigger or schedule data-quality checks for stale availability, missing prices and invalid coordinates.	S	Issues are visible and can notify the business owner.

10.9 Notifications & Communication

ID	Business / Functional Requirement	Priority	Acceptance / Business Rule
FR-080	Users receive transactional email/in-app notifications for booking requests, confirmations, rejections, cancellations, payment status and material itinerary changes.	M	Notifications are triggered by persisted events and are retryable.
FR-081	Businesses receive notifications for new booking requests and time-sensitive actions.	M	Notification includes direct link to the relevant booking.
FR-082	Marketing communication is separate from transactional communication and respects user consent.	M	Users can opt out of marketing without losing essential transactional messages.
FR-083	Notification delivery failures are observable and can be retried through background jobs.	S	Failure does not duplicate booking/payment actions.

11. AI, Recommendation & Optimization Requirements

11.1 AI System Role

The AI subsystem interprets human intent, enriches search, ranks candidates, explains recommendations and helps revise plans. It is not the source of truth for price, business identity, availability, opening hours, route time, payment status or booking status. Those values must come from structured systems and external APIs.

Core AI architecture rule

LLM = language and reasoning interface. Database + external APIs + deterministic services = factual authority. OR-Tools/rules = hard-constraint planner. The model never receives permission to invent a missing business, price, time slot or payment outcome.

11.2 AI Planning Pipeline

Stage	Requirement
1. Intent Extraction	Convert free text into typed fields: start, date, group size, budget, time window, interests, exclusions, travel tolerance, desired pace.
2. Clarification	Ask only for missing mandatory information or resolve ambiguous values.
3. Candidate Retrieval	Query structured inventory using PostGIS, categories, availability and filters. RAG may add verified unstructured context.
4. Eligibility Filtering	Remove impossible candidates: closed, capacity mismatch, excluded category, unreachable, price/budget mismatch.
5. Ranking	Score preference fit, quality, distance, popularity/reliability, novelty and business rules.
6. Constraint Optimization	Use OR-Tools/rules to construct a feasible sequence under time, budget, route and availability constraints.
7. Validation	Recalculate totals and re-check every hard constraint. Reject invalid plan before display.
8. Explanation	LLM writes concise human explanation using only validated itinerary facts.
9. Feedback Capture	Store accepted/replaced/removed items and eventual booking/review outcomes for evaluation and future learning.

11.3 RAG Requirements

- RAG sources are restricted to verified Mshwar content, approved business descriptions/policies and curated destination knowledge.
- Structured data remains authoritative for identifiers, coordinates, price, availability, capacity, opening hours and booking rules.
- Retrieved context must retain source IDs so responses can be traced back to platform records.
- Embeddings are stored in pgvector; re-indexing occurs when source content materially changes.
- The platform must defend against prompt injection in user text or retrieved business content; retrieved text is data, not executable instruction.
- If no trustworthy source supports a factual answer, the assistant should state that the information is unavailable rather than fabricate it.

11.4 Recommendation Scoring

A practical initial ranking function can combine weighted features; weights are configurable and later replaceable by learned ranking. Example signals:

Signal	Examples	Hard vs Soft
Preference Fit	Category, vibe, dietary/family suitability, desired intensity	Soft
Geographic Fit	Distance from route/start, detour cost	Soft + hard max if user sets one
Budget Fit	Per-person/total cost and remaining budget	Hard when user defines strict budget
Time Fit	Opening hours, duration, appointment slot, return deadline	Hard
Availability	Capacity and slot status	Hard for instant book; conditional for request-to-book
Quality / Reliability	Verified rating, cancellation rate, response rate	Soft
Diversity	Avoid repetitive categories; balance trip style	Soft
Sponsored	Optional paid boost within policy	Soft, explicitly labeled and capped

11.5 AI Safety, Reliability & Evaluation

ID	Requirement
AI-01	All LLM structured output must validate against a strict schema before use.
AI-02	No booking, cancellation, refund or price update can be executed directly from unvalidated model text.
AI-03	Recommended entities must exist and be active in the database.
AI-04	Prompt and retrieval content must be bounded to the minimum data needed for the task.
AI-05	Model/provider timeout must fall back to filters/manual planning or deterministic planner where possible.
AI-06	PII and secrets must not be placed into prompts unless required and approved by policy.
AI-07	Evaluation set must include Arabic, Lebanese Arabic, English, French and mixed-language requests.
AI-08	Evaluation must test constraint validity, entity grounding, budget math, route feasibility, language understanding and safe failure.
AI-09	Feedback dataset must store original recommendation, user edits, booking outcome and rating separately from the current production state.
AI-10	AI changes must be versioned so recommendation quality can be compared across prompt/model/ranking versions.

11.6 Feedback Loop & Evaluation Data

The platform should capture the difference between what the system recommended and what the user ultimately chose. This becomes proprietary product data and supports future ranking improvements.

Captured Event	Examples
Original Recommendation	Candidate set, ranking scores, itinerary version, model/ranker version
User Modification	Removed stop, replacement, changed budget, locked stop, changed category
Decision	Saved, shared, abandoned, booked, request sent
Fulfillment	Confirmed, rejected, cancelled, completed, refunded
Outcome	Rating, review, repeat booking, support issue
Context	Trip date, group size, broad location, weather snapshot, route duration; avoid unnecessary PII

12. Business Rules

ID	Business Rule
BR-01	Only active, published entities may appear in new recommendations.
BR-02	Verified badge is controlled by Mshwar admin, never self-assigned by a business.
BR-03	Every user-facing price must have currency, price type and effective source; estimated or from-price values are labeled.
BR-04	Itinerary total is calculated from stored price components; the LLM may explain but not calculate the authoritative total.
BR-05	A strict user budget cannot be exceeded without an explicit warning and user approval.
BR-06	A stop cannot be scheduled outside known opening hours unless the business supports appointment-based exceptions.
BR-07	Instant-confirm inventory requires authoritative capacity control; uncertain inventory must use request-to-book or inquiry.
BR-08	Availability records store last-updated timestamp and source.
BR-09	If availability is considered stale by configurable policy, the UI must communicate uncertainty and may downgrade to request-to-book.
BR-10	Paid/sponsored placement must be clearly labeled and cannot bypass hard eligibility constraints.
BR-11	Reviews must be attributable to a real account and eligibility event; platform may moderate abuse but not rewrite review content.
BR-12	A booking stores immutable snapshots of price and cancellation policy at the time of commitment.
BR-13	Booking/payment operations use idempotency keys to prevent duplicate charges or reservations.
BR-14	Payment success alone does not confirm a booking if the booking state failed; reconciliation must resolve mismatches.
BR-15	Business cancellation requires reason and triggers user notification plus refund/alternative workflow where applicable.
BR-16	User cancellation eligibility follows the booking policy snapshot and applicable law/provider rules.
BR-17	User location is optional; manual start location must always be available.
BR-18	Weather data influences warnings/recommendations but does not guarantee safety or conditions.
BR-19	AI-generated itinerary versions must be persisted before booking so the booked plan is auditable.
BR-20	Admin overrides to financial or booking states require elevated permissions, reason and audit log.
BR-21	Business staff cannot access other businesses' customer or booking data.
BR-22	Data used for personalization is subject to privacy settings and may be reset/deleted according to policy.
BR-23	Platform must not represent unverified scraped data as live availability.
BR-24	A business can pause an experience without deleting historical bookings, reviews or analytics.
BR-25	The platform must preserve a clear distinction between an itinerary suggestion, a reservation request and a confirmed booking.

13. Data Requirements & Core Entities

13.1 Data Categories

Category	Purpose	Examples
Operational Data	Current application state and transactions.	Users, businesses, experiences, availability, bookings, payments, reviews, trips.
Knowledge Data	Verified information used by search/RAG/recommendations.	Descriptions, policies, amenities, destination guides, suitability tags, FAQs.
Learning / Feedback Data	Historical behavior used to improve ranking or evaluate product.	Clicks, saves, replacements, bookings, ratings, business response outcomes.
Evaluation Data	Fixed test cases used to measure AI/recommendation quality over time.	Representative prompts, expected constraints, known feasible itineraries, failure cases.
Audit / Security Data	Accountability and security operations.	Admin actions, business changes, auth events, webhook logs, state transition history.

13.2 Core Data Entities

Entity	Purpose
User	Identity, language, account status and preference references.
UserPreference	Optional interests, exclusions, accessibility, dietary and travel preferences.
Business	Organization, verification, public profile and operational status.
BusinessMember	Staff membership and role.
Destination	City/area/region used for discovery and content.
Place / Venue	Physical location with coordinates and address.
Experience	Bookable or discoverable offering with category, duration, price and suitability.
OperatingHours	Regular opening schedule and exceptions.
AvailabilitySlot	Date/time, capacity, source and update timestamp.
PriceRule	Currency, price type, amount/range, unit and effective dates.
Trip	User/group itinerary container and planning metadata.
TripVersion	Immutable/generated version for edits and audit.
TripStop	Ordered experience/place with scheduled time, duration and cost.
GroupParticipant	Shared-trip membership, role and optional shared preferences.
Vote	Group vote on experience/category/option.
Booking	Reservation lifecycle and snapshot of price/policy.
Payment	Provider-neutral transaction record and external provider reference.
Refund	Refund request/result linked to payment/booking.
Review	Rating, text, moderation state and eligibility reference.
Favorite	User-to-listing/trip save relationship.
Notification	Transactional communication state.
KnowledgeDocument	RAG-indexable verified content and source metadata.
RecommendationEvent	Candidate/ranking/itinerary output with version metadata.
FeedbackEvent	User modifications, acceptance, conversion and outcome.
AuditLog	Actor, action, target, timestamp and metadata for sensitive changes.

13.3 Geospatial Data

- Store canonical latitude/longitude and PostGIS geography/geometry for venues and relevant destinations.
- Support radius, nearest-neighbor, bounding-box and route corridor queries.
- Do not use geocoded text alone as the primary location key; persist normalized coordinates plus source/provider ID where permitted.
- Respect external map provider terms on storage/caching of derived data.

14. External Integrations

Integration	Interface	Purpose	Key Control
Google Maps Platform	Maps JavaScript API, Places/Geocoding as required, Routes API	Map display, place resolution, route distance/time	Quota/cost, API key restrictions, provider terms
Open-Meteo	Forecast API	Weather-aware warnings/replanning	Free endpoint is non-commercial; commercial/self-hosted production plan required
ImageKit	Media upload/delivery API	Business/experience images and transformations	Upload validation, signed/authenticated uploads, plan limits
Stripe	Checkout/Payment Intents/Webhooks in test environment	Payment demo and provider abstraction validation	Lebanon merchant availability constraint; production PSP decision required
Resend	Email API	Transactional email	Deliverability, domain verification, rate limits
Sentry	SDK/API	Error/performance monitoring	PII scrubbing and sampling
PostHog	Event analytics	Product funnels and behavior analytics	Consent, PII minimization, event governance
LLM Provider / Ollama	Model API	Intent extraction, explanation, conversational edits	Cost/latency/model availability; provider abstraction

15. Non-Functional Requirements

ID	Quality Attribute	Requirement
NFR-01	Performance	Core browse pages target p75 LCP <=2.5s where feasible; API endpoints should generally respond <500ms excluding external AI/maps calls.
NFR-02	AI Latency	Trip generation should stream/progress feedback and target useful first result within 10-15 seconds under normal provider conditions.
NFR-03	Availability	MVP target 99.5% monthly availability excluding planned maintenance and third-party outages.
NFR-04	Scalability	Stateless app/API services should scale horizontally; background jobs decoupled via queue.
NFR-05	Database Integrity	Bookings/payments use transactions, constraints and idempotency; referential integrity enforced.
NFR-06	Security	OWASP-aligned controls, secure headers, validation, least privilege, secret management, rate limiting and audit logs.
NFR-07	Privacy	Data minimization, purpose limitation, retention rules, consent for optional personalization/location.
NFR-08	Localization	Arabic/English/French content and RTL support; dates/currency/time formatted consistently.
NFR-09	Accessibility	Target WCAG 2.1 AA for key user journeys; keyboard navigation and semantic labels.
NFR-10	Mobile	Responsive layouts optimized for common mobile widths and touch targets.
NFR-11	Observability	Structured logs, correlation IDs, Sentry errors, metrics and health endpoints.
NFR-12	Resilience	Third-party failures degrade gracefully; user action is not lost silently.
NFR-13	Backup/Recovery	Automated database backups and tested restoration process; RPO/RTO defined before production launch.
NFR-14	Maintainability	Typed interfaces, migrations, automated tests, linting, code review and documented environments.
NFR-15	Compatibility	Support current major Chrome, Safari, Edge and Firefox versions; mobile Safari/Chrome prioritized.
NFR-16	SEO	Public destination/listing pages use server rendering/metadata, canonical URLs and structured content where appropriate.
NFR-17	Data Freshness	Availability, price and policy freshness are observable; stale records are flagged or downgraded.
NFR-18	Auditability	Sensitive admin/business/payment state changes record actor, timestamp, reason and old/new state where practical.
NFR-19	Cost Control	External API quotas, alerts and caching strategies prevent uncontrolled spend.

ID	Quality Attribute	Requirement
NFR-20	AI Grounding	No user-visible recommended business may originate only from model memory; all must map to platform records.

16. Security, Privacy & Trust

16.1 Identity & Access

- Role-based access control for traveler, business owner, business staff, support/admin and super-admin responsibilities.
- Administrative actions should use stronger authentication controls before production, including MFA for privileged accounts.
- Session/token validation occurs server-side; authorization checks are not delegated to the frontend.
- Organization membership is enforced on every business data access path.

16.2 Application Security

- Validate all request bodies with Pydantic and enforce database constraints.
- Rate-limit authentication, AI generation, search abuse, booking creation and sensitive admin endpoints.
- Use secure secret storage; no API keys in frontend bundles or Git repositories.
- Restrict Google Maps keys by domain/API and use server-side keys for server-only APIs.
- Verify payment webhook signatures and use idempotency keys for financial actions.
- Validate image MIME/type/size and use controlled ImageKit upload flow.
- Protect against XSS, CSRF where applicable, SQL injection, SSRF, broken object authorization and mass assignment.
- Sanitize/escape user-generated content and apply moderation rules to reviews/business descriptions.

16.3 AI-Specific Security

- Treat user prompts and RAG documents as untrusted data; they cannot redefine system permissions.
- LLM tools/functions are allow-listed and backend-mediated. High-impact actions require explicit user confirmation and deterministic checks.
- Prompt payloads exclude secrets and unnecessary personal data.
- Model output is schema-validated, entity-grounded and rechecked before affecting itinerary or booking state.

16.4 Trust & Marketplace Safety

- Business verification workflow with evidence retention appropriate to policy.
- Report/flag mechanism for listings, reviews and safety concerns.
- Clear distinction between verified, unverified, sponsored and user-generated content.
- Cancellation, refund, pricing and availability policies shown before booking confirmation.

17. Payments, Booking, Refunds & Commercial Constraints

17.1 Booking Modes

Mode	Behavior	Use Case
Instant Confirm	Capacity is reserved and booking is confirmed immediately after successful required steps/ payment.	Businesses maintaining authoritative slot inventory.
Request to Book	User requests a slot; business must confirm within configured SLA before final commitment.	Businesses with manually maintained or uncertain availability.
Inquiry Only	Mshwar captures qualified lead/contact request; no platform booking guarantee.	Early supply categories or quote-based experiences.

17.2 Payment Architecture

Payments must use a provider-neutral service boundary. Stripe can be used in test mode for the university/demo implementation. The core Booking and Payment entities must not assume Stripe-specific fields as primary identifiers.

Production Lebanon constraint

As of 10 September 2026, Lebanon is not listed on Stripe's official global availability page for direct account support. Therefore, a real Lebanon launch must select a legally available payment/acquiring solution or operate through an eligible supported business structure. The application must keep payment-provider abstraction so this decision does not require redesigning bookings.

17.3 Refund & Reconciliation Requirements

- Refund amount is derived from persisted policy and payment data, not free-form AI output.
- Webhook and client callback states are reconciled; the server does not trust a browser redirect as proof of payment.
- Partial refund must be represented separately from full refund.
- Operational reconciliation report identifies paid-but-unconfirmed, confirmed-but-unpaid and webhook-failure anomalies.
- Marketplace payout/split-payment design is deferred until the legally available production payment model is confirmed.

18. Failure Handling & Fallback Design

Failure	Required Fallback
LLM unavailable/timeout	Fall back to structured filters and deterministic planner; allow manual trip building; show concise retry option.
RAG unavailable	Use structured listing fields only; suppress unsupported narrative claims.
Google Maps route failure	Show stops/list without authoritative route time; do not pretend optimized route is valid.
Weather unavailable	Keep itinerary but show weather data unavailable; do not trigger weather-based replacement.
Business availability stale	Downgrade to request-to-book/inquiry and display uncertainty.
Payment provider failure	Keep booking unconfirmed/pending as appropriate; no duplicate payment retries without idempotency.
Email failure	Persist notification and retry asynchronously; user can still see status in-app.
Image service failure	Use safe placeholder; do not block core booking flow.
Optimization finds no feasible plan	Explain which constraints conflict and offer actionable relaxations: budget, distance, time or category.
Business cancellation	Notify user, process refund if due and suggest valid replacements.

19. Analytics, Reporting & Operational Controls

19.1 Product Analytics

Funnel Area	Example Events
Acquisition	landing_view, destination_view, listing_view, search_submitted
Planning	trip_builder_started, intent_parsed, clarification_shown, itinerary_generated, itinerary_failed
Editing	stop_locked, stop_removed, stop_replaced, budget_changed, weather_replan
Engagement	favorite_added, trip_saved, trip_shared, vote_cast
Conversion	booking_started, booking_requested, booking_confirmed, payment_started, payment_succeeded
Fulfillment	booking_completed, booking_cancelled, refund_created, review_submitted
Supply	business_verified, experience_published, slot_updated, booking_response

19.2 Business Dashboard Metrics

- Profile/listing views
- Search impressions and itinerary inclusions
- Favorites/saves
- Booking requests

- Confirmation/rejection/cancellation rate
- Revenue/booking value where available
- Average response time
- Rating/review trends

19.3 Operational Controls

- Feature flags for AI models, ranking versions, payments and new marketplace features.
- External API budget/quota alerts.
- Stale data dashboard for listings, prices and availability.
- Booking/payment reconciliation dashboard.
- Admin audit logs and moderation queue.
- Error dashboards in Sentry plus infrastructure/service health checks.

20. MVP Acceptance Criteria

ID	MVP Acceptance Criterion
AC-01	A user can create an account, select a language and maintain preferences.
AC-02	A user can browse real seeded businesses/experiences and view them on a Google Map.
AC-03	A user can submit a natural-language trip request and receive a structured, feasible itinerary using only seeded/verified entities.
AC-04	Trip generation respects strict budget, opening hours, duration, group size and return-time constraints, or explicitly reports that no feasible plan exists.
AC-05	The system calculates routes/travel times and sequences a multi-stop plan.
AC-06	Weather-sensitive stops display a weather warning when configured conditions are met and can be replanned.
AC-07	A user can replace/lock stops and the itinerary recalculates time, route and cost.
AC-08	A user can share a trip and collect participant votes.
AC-09	A business can create/manage an experience, pricing and availability and receive booking requests.
AC-10	An admin can verify a business and moderate listing/review content.
AC-11	The booking flow supports pending/confirmed/cancelled states with policy/price snapshots.
AC-12	Stripe test-mode payment can be demonstrated without exposing secret keys, and payment webhooks are verified.
AC-13	AI provider failure has a usable fallback path and does not corrupt persisted state.
AC-14	Automated tests cover critical auth, trip-generation validation, booking and payment state transitions.
AC-15	Sentry captures controlled test errors in deployed environment; logs contain correlation IDs and no secrets.
AC-16	Arabic pages render RTL correctly; English/French render LTR; Lebanese Arabic prompts are included in evaluation tests.

21. Delivery Phases & Roadmap

Phase	Objective	Key Deliverables
Phase 0 - Foundation	Prove architecture and data model.	Repo/CI, design system, auth, PostgreSQL/PostGIS, core entities, seed/admin tools, environments.
Phase 1 - Marketplace MVP	Enable discovery and trusted inventory.	Public listings, search/filter/map, business portal, verification, images, favorites.
Phase 2 - AI Planner	Turn intent into validated itineraries.	Intent extraction, structured retrieval, ranking, OR-Tools, route/time, itinerary editor, RAG explanations.
Phase 3 - Booking	Convert plans to transactions.	Availability, request-to-book, booking states, notifications, Stripe test integration, reconciliation.
Phase 4 - Group + Weather	Differentiate planning experience.	Shared trips, voting, group fit, weather warnings and partial replanning.
Phase 5 - Production Pilot	Operate with real businesses.	Commercial payment decision, legal/policies, SLA/support, monitoring, analytics, partner onboarding, security review.
Future	Scale marketplace and intelligence.	Learned ranking, deeper personalization, supplier APIs, loyalty, native apps, transport/partner integrations.

22. Risks, Assumptions & Dependencies

22.1 Key Risks & Mitigations

ID	Risk	Impact	Mitigation	Severity
R-01	Supply/data freshness	Businesses may not update availability/pricing.	Freshness timestamps, reminders, request-to-book fallback, admin quality checks.	High
R-02	Cold-start marketplace	Too few businesses reduces itinerary quality.	Curated launch inventory, ready-made trips, focus on dense categories/areas before broad expansion.	High
R-03	AI hallucination	Model invents facts or ignores constraints.	Structured retrieval, schema validation, entity grounding, deterministic checks, fixed eval set.	High
R-04	Payments in Lebanon	Selected PSP may not support local merchant account.	Provider abstraction; validate local/regional acquiring before production launch.	High
R-05	Third-party API cost	Maps/AI/weather usage may grow unexpectedly.	Caching, quotas, budget alerts, provider abstraction, self-host options where appropriate.	Medium
R-06	Business cancellation/poor service	Damages user trust.	Reliability metrics, verified partners, clear policies, support/refund/alternative flow.	High
R-07	Privacy/security	Account, location or payment data exposure.	Least privilege, data minimization, secure secrets, audits, testing and monitoring.	High
R-08	Over-scoped MVP	Too many features delay a usable product.	Phase gates; preserve core discovery -> plan -> validate -> book flow first.	High
R-09	Weather uncertainty	Forecast can be wrong.	Present as advisory, show timestamp/source, never guarantee conditions.	
R-10	Routing dependency	Map provider outage or terms/cost changes.	Graceful degradation, cache allowed data, provider abstraction at service layer where practical.	Medium

22.2 Assumptions

- Participating businesses will provide or approve accurate public listing, pricing and policy data.
- Mshwar can obtain suitable map/routing API access and production budget.
- Users have internet access and modern mobile/desktop browsers.
- A legally available payment solution can be selected before commercial checkout is enabled in Lebanon.
- The platform will define terms, privacy, cancellation/refund and marketplace policies before production launch.
- The academic/demo environment can use synthetic/test payments and seeded business data without representing them as real transactions.

22.3 Dependencies

- Verified business onboarding and content operations.
- Google Maps Platform APIs and quotas.
- Weather provider licensing/availability.
- LLM provider/local compute capacity.
- ImageKit and email delivery services.
- Payment provider/legal entity/acquirer decision for commercial launch.

23. Technology Baseline

The technology list is a baseline, not a requirement to use every tool on day one. Avoid duplicate tools that solve the same responsibility. Final choices should be recorded in an Architecture Decision Record (ADR).

Layer	Technology
Frontend	Next.js, React, TypeScript, Tailwind CSS, shadcn/ui
Backend	FastAPI, Python, SQLAlchemy, Pydantic, Alembic
Database	PostgreSQL, PostGIS, pgvector (initially hosted on Supabase if suitable)
Authentication	Auth.js or Better Auth; choose one during technical design and avoid duplicate auth systems
AI / RAG	LLM provider abstraction: OpenAI API and/or Ollama for local development; pgvector RAG; Hugging Face models where useful
Recommendation / ML	scikit-learn, NumPy/Pandas as needed
Optimization	Google OR-Tools
Maps & Location	Google Maps Platform: Maps JavaScript API, Places/Geocoding as required, Routes API
Weather	Open-Meteo prototype; commercial plan or self-hosted production option
Images	ImageKit
Payments	Stripe test mode + provider-neutral payment service
Caching / Jobs	Redis, Celery
Real-Time	WebSockets for selected live updates; polling/event-driven fallback acceptable
Email	Resend
Monitoring	Sentry
Analytics	PostHog
Testing	Pytest, Vitest, React Testing Library, Playwright
Code Quality	ESLint, Prettier, Ruff, mypy
DevOps	Docker, Docker Compose, Git, GitHub, GitHub Actions
Deployment	Vercel frontend; Supabase/PostgreSQL data; backend runtime chosen based on FastAPI/Celery/Redis requirements

23.1 Logical Solution Context



24. Testing & Quality Strategy

Test Layer	Scope
Unit Tests	Pricing math, constraint rules, availability logic, state transitions, schema parsing, ranking utilities.
API/Integration Tests	Auth, database constraints, maps/weather adapters, ImageKit, email, payment webhooks using test doubles/sandboxes.
End-to-End Tests	Browse -> plan -> edit -> book; business confirm; cancel/refund; group vote; admin verify.
AI Evaluation	Fixed multilingual prompts, entity-grounding checks, budget/time validity, no-result cases, adversarial prompt injection cases.
Performance Tests	Search, trip generation concurrency, booking contention, queue throughput.
Security Tests	Authorization, rate limits, webhook verification, common OWASP cases, secret exposure checks.
Accessibility Tests	Automated checks plus manual keyboard/screen-reader validation on core flows.
UAT	Realistic user and business scenarios against acceptance criteria with representative seeded data.

24.1 Critical Test Scenarios

- Budget too low for any feasible itinerary.
- User requests places that are closed on the selected day.
- Two users attempt the final remaining slot concurrently.
- LLM returns malformed JSON or an unknown entity ID.
- Map API times out mid-generation.
- Weather changes after a trip has been saved.
- Payment webhook arrives more than once or before browser callback.
- Business cancels after payment.
- Arabic/mixed-language request contains ambiguous numbers or currency.

- Prompt injection text is included inside a business description retrieved by RAG.

25. Deployment, DevOps & Observability

25.1 Environments

Environment	Purpose	Data
Local	Developer implementation and unit/integration tests.	Synthetic/local seed data; local Ollama optional.
Preview	Pull-request/branch UI and API validation.	Isolated non-production data/services.
Staging	UAT, E2E, payment test mode, production-like integration tests.	Sanitized/seeded data; no live financial transactions.
Production	Real users/businesses.	Live data, production secrets, backups, monitoring, legal/payment configuration.

25.2 CI/CD Controls

- GitHub Actions runs lint, type checks, tests and build before merge/deploy.
- Database migrations are versioned and tested; destructive changes require explicit review/backout plan.
- Secrets are stored in deployment secret managers, never in repository or client code.
- Production deploys use protected branch/environment approvals as the project matures.
- Sentry release tags link errors to deployed commits; health checks validate critical services after deploy.

25.3 Observability

- Structured server logs with request/correlation IDs.
- Sentry for frontend/backend exceptions and selected performance traces.
- Metrics for API latency/errors, queue backlog, AI provider latency, booking failures and external API failures.
- Alerting for payment reconciliation anomalies, elevated error rate, database/storage issues and external quota exhaustion.
- PII/secrets scrubbed from logs and monitoring payloads.

26. External Service Constraints

The following checks are included because Mshwar is intended to become a real commercial product, not only a classroom demo. Commercial terms can change and must be re-verified before launch.

Service	Verified Constraint (10 Sep 2026)	BRD Impact
Stripe	Stripe's official global availability list does not list Lebanon as a directly supported country/region.	Use test mode for demo; keep production payment provider abstract and confirm a legally available acquirer/PSP before launch.
Google Maps Platform	Core services use pay-as-you-go pricing with SKU-specific free monthly usage caps and volume pricing.	Set quotas/budget alerts, minimize unnecessary requests and model maps cost in production economics.
Open-Meteo	Free API is for non-commercial use; commercial use requires a paid commercial licence/API or self-hosting. Attribution applies to data.	Free endpoint is appropriate for prototype/academic use; production commercial use must change plan or self-host and preserve attribution.

26.1 Source References

- Stripe Global Availability - <https://stripe.com/global> (checked 10 September 2026)
- Google Maps Platform Pricing - <https://developers.google.com/maps/billing-and-pricing/pricing> (checked 10 September 2026)
- Open-Meteo Pricing - <https://open-meteo.com/en/pricing> (checked 10 September 2026)
- Open-Meteo Terms - <https://open-meteo.com/en/terms> (checked 10 September 2026)

27. Glossary & Appendices

Term	Meaning
BRD	Business Requirements Document.
RAG	Retrieval-Augmented Generation: providing a model with retrieved, approved source content at request time.
LLM	Large Language Model used for language interpretation/generation.
PostGIS	PostgreSQL geospatial extension for location and distance queries.
pgvector	PostgreSQL extension for vector embeddings/similarity search.
OR-Tools	Optimization toolkit used for sequencing and constraint-solving.
Request-to-Book	Reservation request requiring business confirmation before becoming confirmed.
Instant Confirm	Booking mode where authoritative inventory allows immediate confirmation.
Hard Constraint	Condition that must be satisfied, e.g. return time, capacity or strict budget.
Soft Preference	Preference used for ranking but may be traded off, e.g. vibe or novelty.
Grounding	Ensuring AI claims/recommendations are tied to trusted platform sources.
PSP	Payment Service Provider.
RPO/RTO	Recovery Point Objective / Recovery Time Objective.

27.1 Requirement Priority Legend

Priority	Meaning
Must (M)	Required for MVP acceptance or core platform integrity.
Should (S)	High-value requirement that may move if schedule demands without breaking the core product.
Could (C)	Useful enhancement that is not required for initial acceptance.

27.2 Definition of Done for the BRD Baseline

- Product owner accepts product vision, scope and marketplace model.
- MVP Must requirements are approved or explicitly changed.
- Payment constraint and commercial-weather licensing constraint are acknowledged.
- Key business rules and acceptance criteria are used to produce FRD/SRS and UX flows.

- Architecture team records final technical choices in ADRs rather than silently diverging from the baseline.